Bayesian Machine Learning

Presentation - Project Assignment

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The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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Bayesian Computation and MCMC

- For complex models, exact posterior inference is often intractable
- MCMC methods are asymptotically unbiased and broadly applicable [Nea93]
- Traditional algorithms (random-walk Metropolis [Met+53], Gibbs sampling [GG84]) suffer from inefficient exploration in high-dimensional or highly correlated parameter spaces

Goal: Review and implement the No-U-Turn Sampler (NUTS) [HG+14], an extension of Hamiltonian Monte Carlo that eliminates manual path-length tuning

Hamiltonian Monte Carlo (HMC)

HMC [Nea+11; Dua+87] introduces auxiliary momentum variables $r \in \mathbb{R}^D$ paired with parameters $\theta \in \mathbb{R}^D$:

$$p(\theta, r) \propto \exp\left\{\mathcal{L}(\theta) - \frac{1}{2} r \cdot r\right\}$$

Each iteration:

1. Resample $r \sim \mathcal{N}(0, I)$
2. Simulate Hamiltonian dynamics via leapfrog integrator for L steps of size ε
3. Accept/reject via Metropolis step

Volume-preserving proposals can be simultaneously **distant** and **highly probable**.

Limitations of HMC

Performance depends critically on two parameters:

- **Step size ε:** Too large $\Rightarrow$ high rejection; too small $\Rightarrow$ wasted computation
- **Number of leapfrog steps L:** Too small $\Rightarrow$ random walk; too large $\Rightarrow$ trajectory loops back, wasting gradient evaluations

Selecting a good L typically requires costly preliminary runs and considerable expertise, limiting HMC's adoption in general-purpose inference software.

NUTS addresses both issues:

- Adaptive trajectory length via the No-U-Turn criterion
- Adaptive step size via dual averaging [Nes09]

The No-U-Turn Criterion

Key idea: Stop trajectory simulation when further steps would move the proposal *back* towards the starting point.

Monitor the time derivative of $\frac{1}{2}\|\tilde{\theta} - \theta\|^2$: when this becomes negative, the trajectory is doubling back.

U-turn condition on leftmost/rightmost leaves (θ^-, r^-), (θ^+, r^+):

$$(\theta^+ - \theta^-) \cdot r^- < 0 \quad \text{or} \quad (\theta^+ - \theta^-) \cdot r^+ < 0$$

A naive stopping rule would violate time-reversibility $\Rightarrow$ NUTS builds trajectories **both forwards and backwards** via recursive doubling.

Recursive Doubling and Trajectory Construction

At each doubling step j:

1. Draw direction $v_j \sim \mathrm{Uniform}(\{-1, +1\})$
2. Take 2^j leapfrog steps of size $v_j \varepsilon$
3. Implicitly construct a balanced binary tree of visited states

Slice variable augmentation: $u \sim \mathrm{Uniform}\big([0, \exp\{\mathcal{L}(\theta) - \frac{1}{2} r \cdot r\}]\big)$

Candidate set $\mathcal{C} \subseteq \mathcal{B}$: states lying within the slice.

Doubling halts when:

- A U-turn is detected in any balanced subtree, or
- Numerical instability is encountered

States added during a halting iteration are excluded from $\mathcal{C}$ to preserve detailed balance.

Efficient NUTS: Memory-Efficient Transition Kernel

Problem: Storing all candidate states requires $\mathcal{O}(2^j)$ memory.

Solution: Partition $\mathcal{C}$ into $\mathcal{C}^{\text{old}}$ and $\mathcal{C}^{\text{new}}$ at each doubling.

For $w \in \mathcal{C}^{\text{old}}$, propose $w' \in \mathcal{C}^{\text{new}}$ uniformly, accept with probability:

$$\min\left(1,\ \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right)$$

This reduces memory from $\mathcal{O}(2^j)$ to $\mathcal{O}(j)$ while preserving detailed balance with respect to the uniform distribution over $\mathcal{C}$.

Adaptive Step Size via Dual Averaging

Adapt ε during warm-up to drive the average acceptance statistic towards target δ:

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma(t+t_0)} \sum_{i=1}^{t} H_i, \qquad \bar{x}_{t+1} = t^{-\kappa} x_{t+1} + (1 - t^{-\kappa}) \bar{x}_t$$

where $x = \log \varepsilon$, $\mu = \log(10\varepsilon_0)$, and hyperparameters:

- $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$

Combined with the No-U-Turn criterion, this renders NUTS **entirely tuning-free**.

Our Implementation

From-scratch Python implementation of HMC and NUTS with dual averaging.

Samplers:

- `DualAveragingHMC`
- `NaiveNUTS`
- `EfficientNUTS`
- `DualAveragingNUTS`

Metrics:

- Autocorrelation
- Effective Sample Size (ESS)
- ESS per gradient evaluation

Performance is measured as **minimum ESS across all dimensions** per gradient evaluation.

Benchmark 1: Multivariate Normal (250-D)

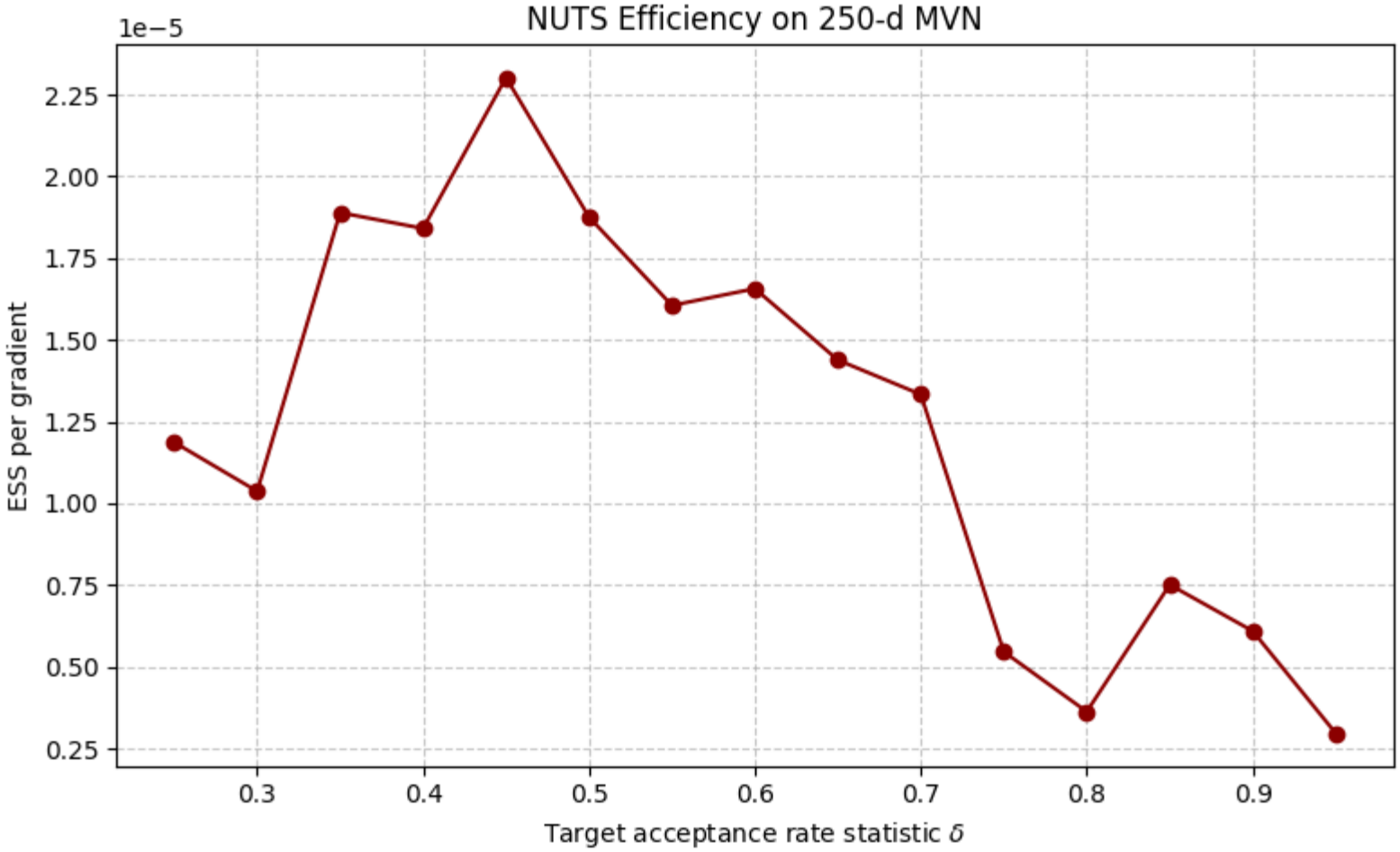

Figure: ESS/gradient for NUTS vs. best-tuned HMC on a 250-D Gaussian with random Wishart precision matrix. NUTS matches or exceeds HMC by roughly a factor of two without trajectory-length tuning.

Benchmark 2: Bayesian Logistic Regression (25-D)

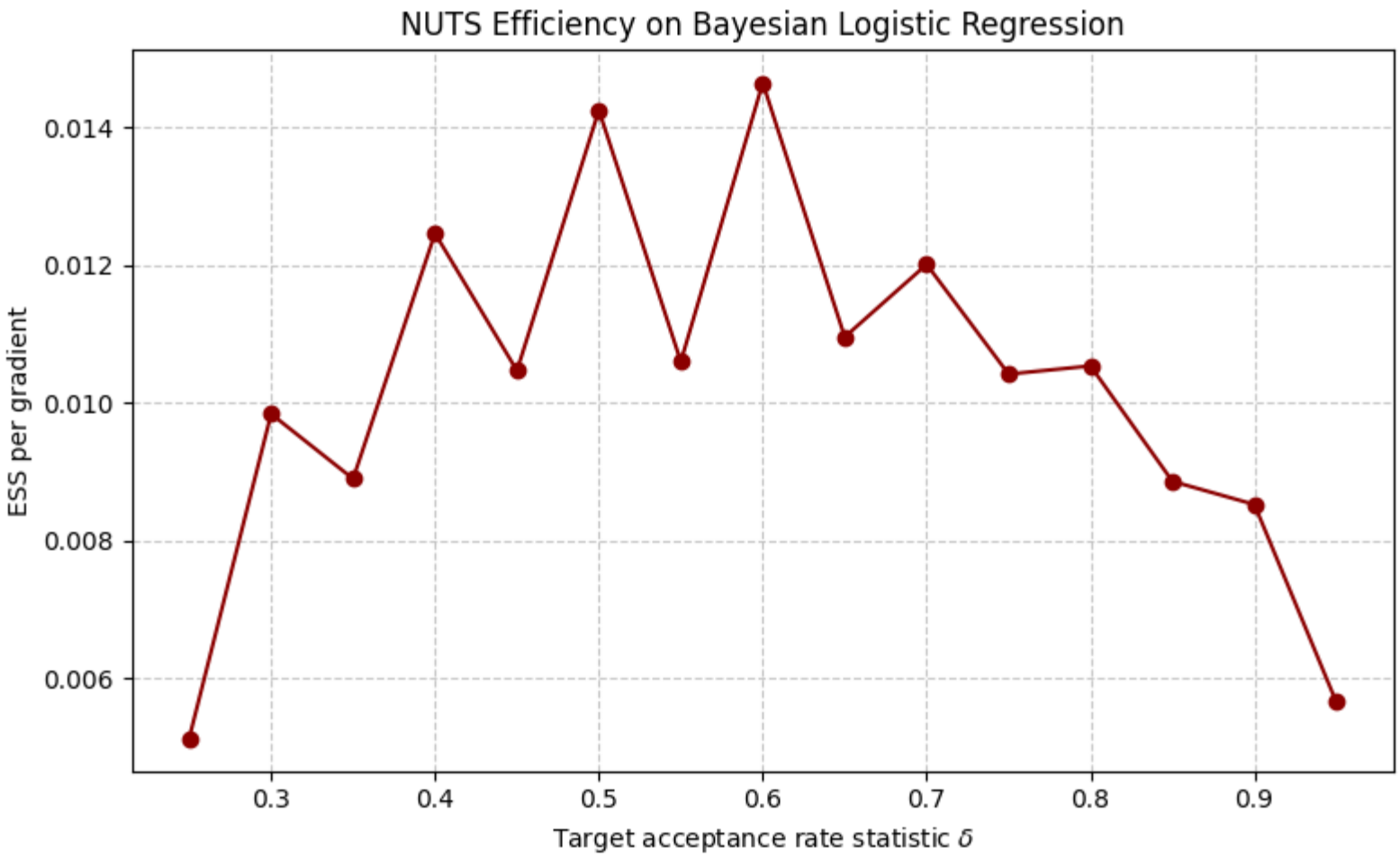

Figure: ESS/gradient on Bayesian Logistic Regression fitted to the German Credit dataset (25 parameters). NUTS achieves comparable efficiency to the best-tuned HMC.

Benchmark 3: Stochastic Volatility (3001-D)

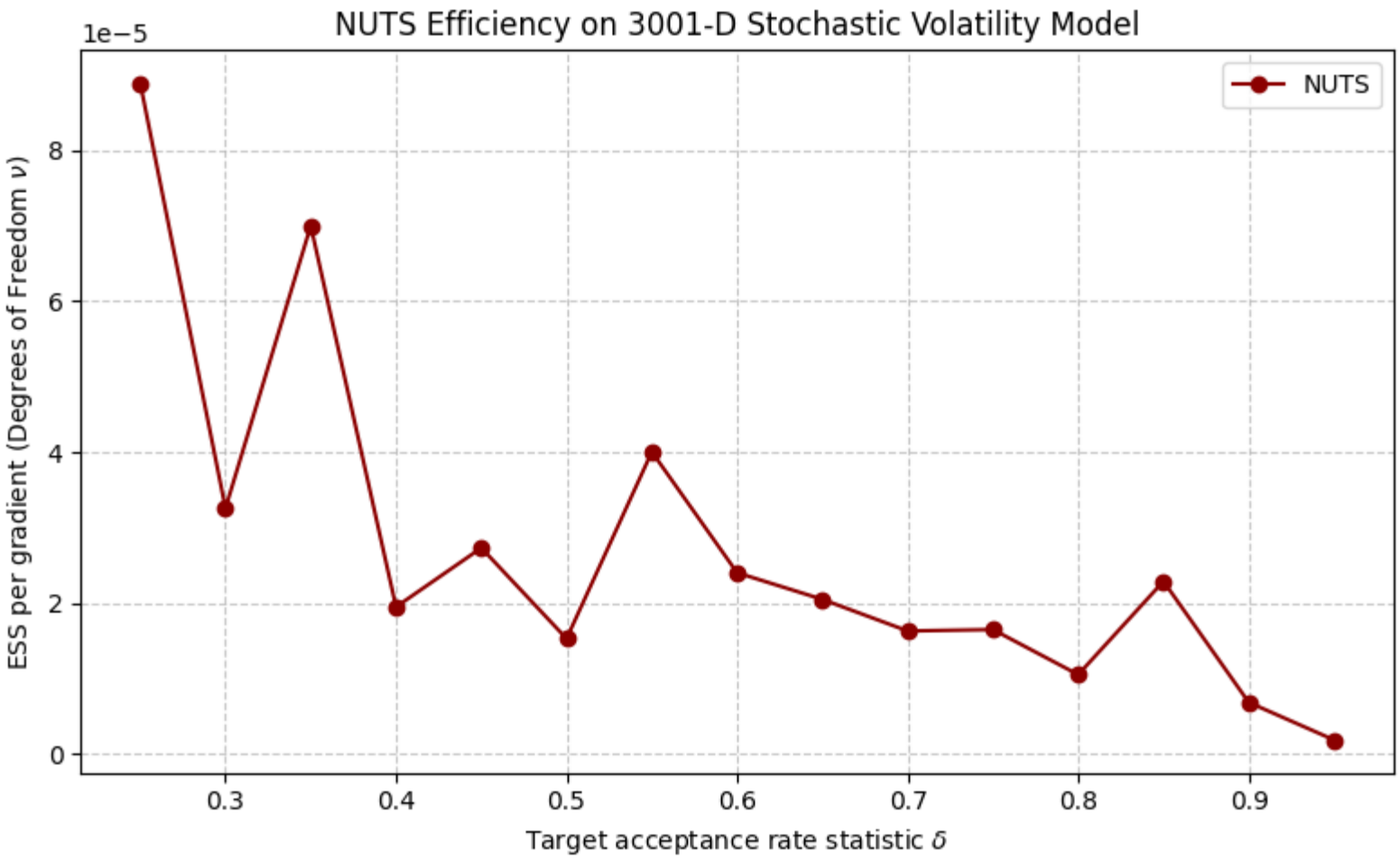

Figure: ESS/gradient on the Stochastic Volatility model fitted to S&P 500 returns (3001-D). NUTS outperforms HMC by roughly a factor of 1.5 without requiring any tuning.

Dual Averaging Convergence

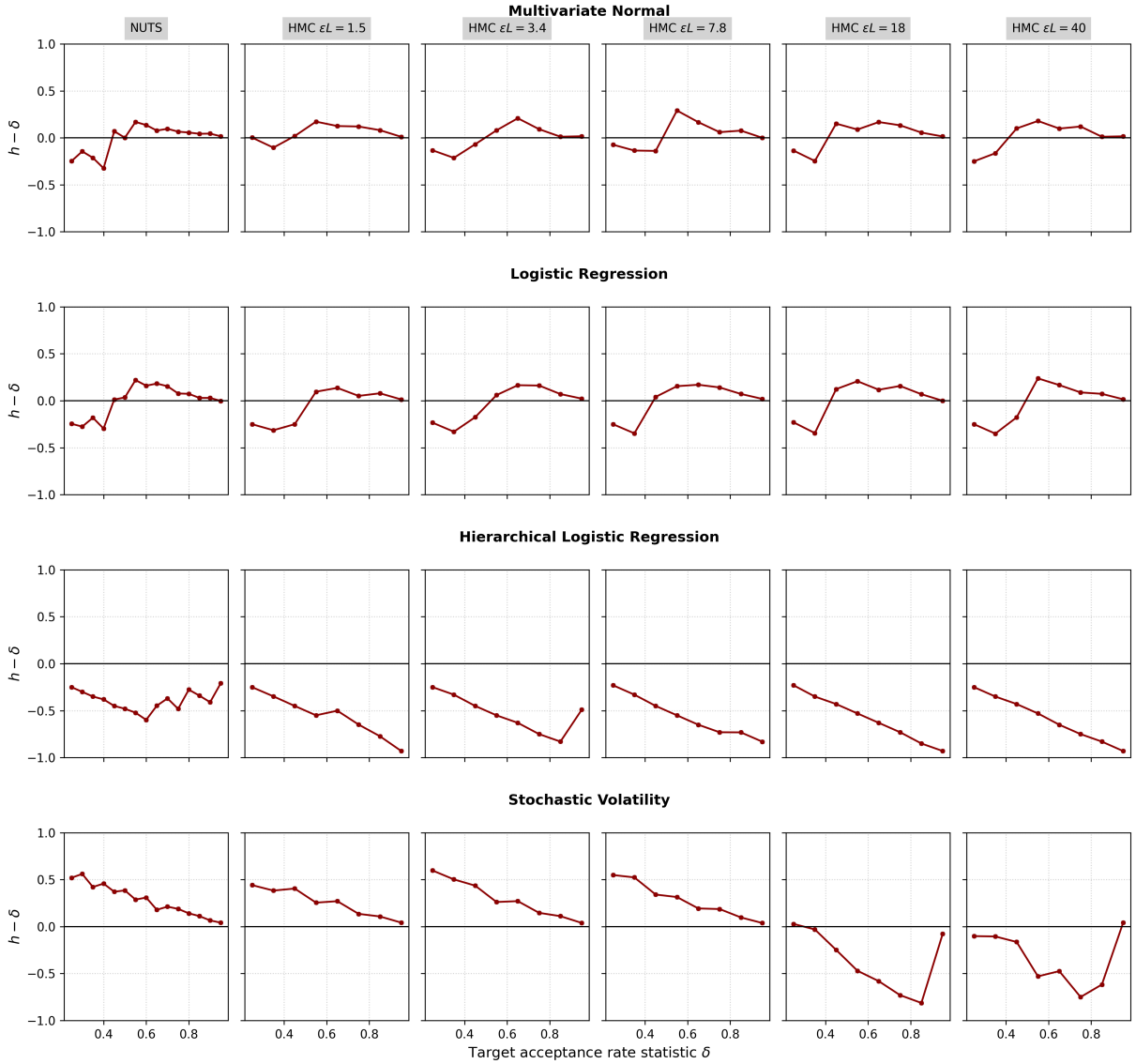

Figure: Discrepancy plots ($h - \delta$ across δ) for all four original benchmarks. The dual-averaging scheme reliably drives the acceptance statistic towards the target δ within a few hundred warm-up iterations.

Motivation

The original evaluation is limited to **unimodal or mildly structured** targets.

We design three experiments targeting pathologies common in modern Bayesian models:

1. **Extreme multimodality** — Bayesian Neural Network
2. **Dense correlations** — Gaussian Process Classification
3. **Spatially varying geometry** — Neal's Funnel

These stress tests probe failure modes of HMC that are **unexplored in the original paper**.

Stress Test 1: Bayesian Neural Network (191-D)

Setup:

- Two-hidden-layer BNN ($30 \to 5 \to 5 \to 1$), 191 parameters
- $\mathcal{N}(0, 100)$ priors on all weights and biases
- UCI Breast Cancer dataset
- Weight-permutation symmetries create $5! \times 5! = 14{,}400$ equivalent modes

Configuration: $\delta = 0.65$, $M = 300$, $M_{\text{adapt}} = 150$

NUTS adapts trajectory lengths between 1 and 64 leapfrog steps, traversing curved valleys connecting symmetric modes. HMC, constrained to a fixed trajectory, overshoots or undershoots and stalls in a single mode.

BNN: Trajectory Lengths and Efficiency

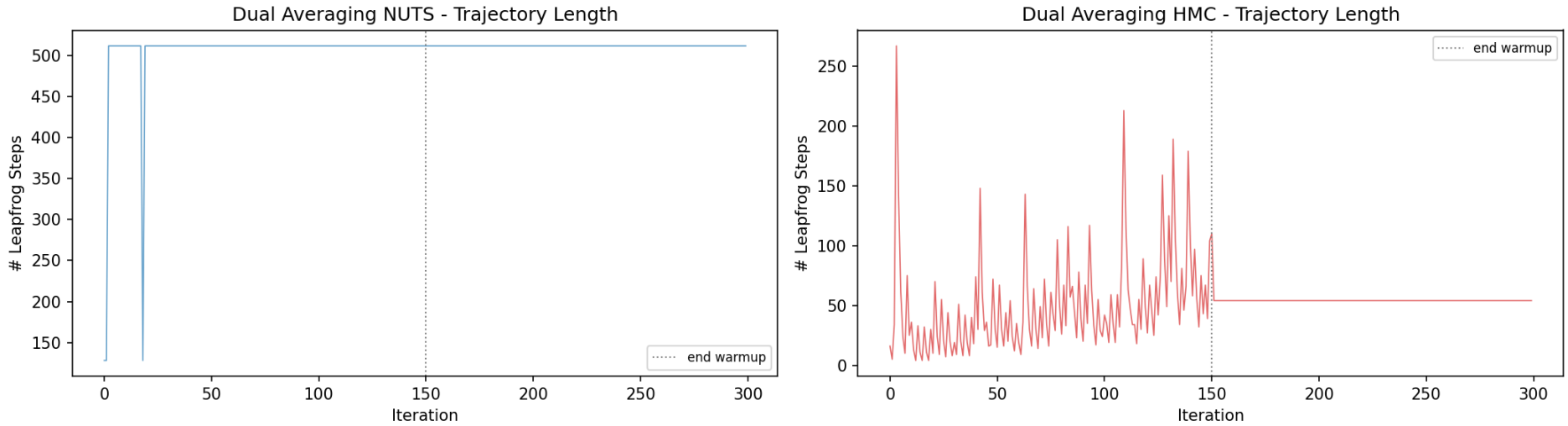

Figure: NUTS trajectory lengths on the BNN posterior. Lengths concentrate on powers of two, confirming the U-turn criterion triggers at complete doublings. The adaptive mechanism is essential for navigating the multimodal landscape.

BNN: Trace Plots

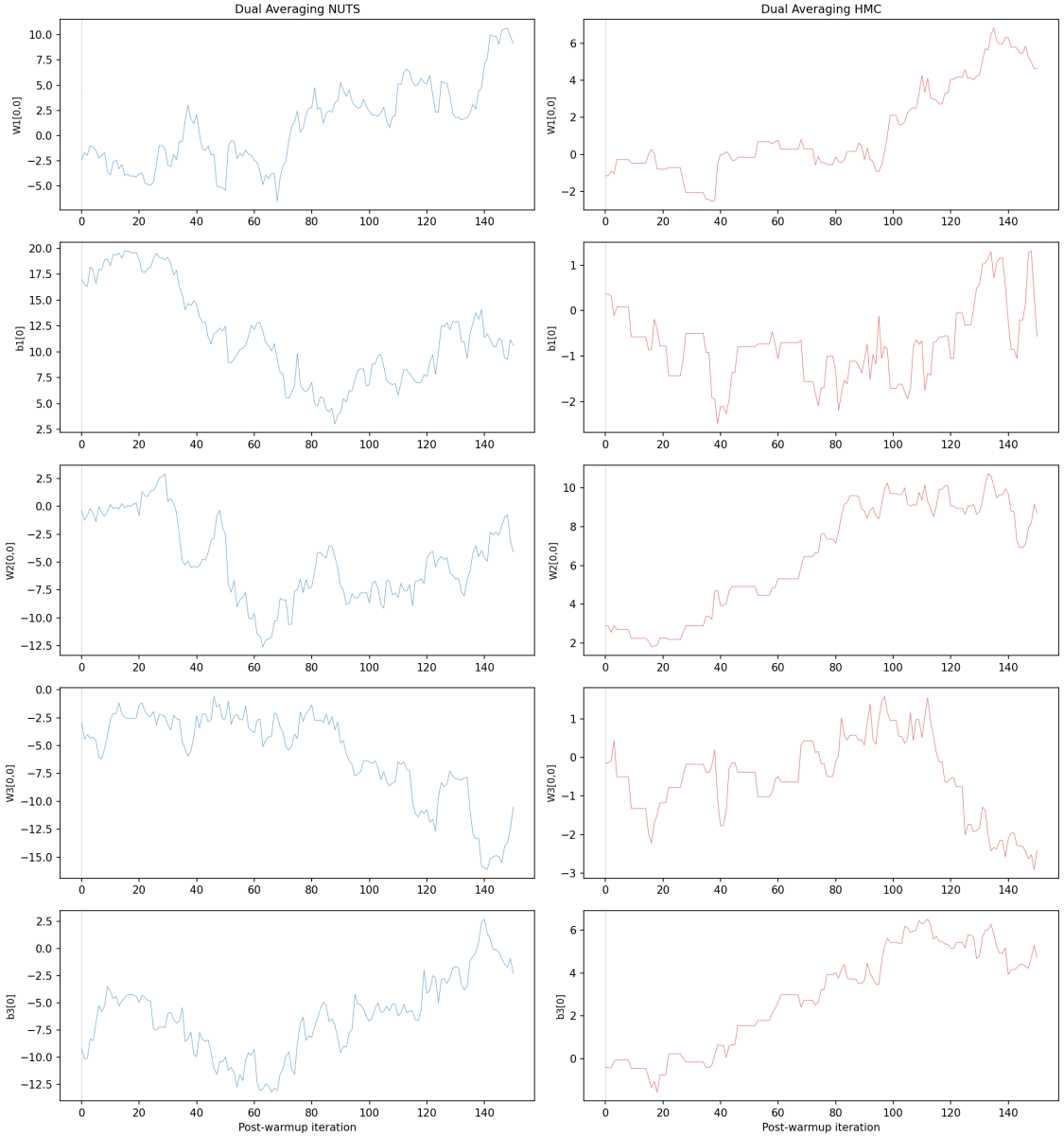

Figure: Trace plots on the BNN posterior. NUTS explores a wider range of weight values, while HMC exhibits prolonged stagnation in a single mode.

BNN: Efficiency Comparison

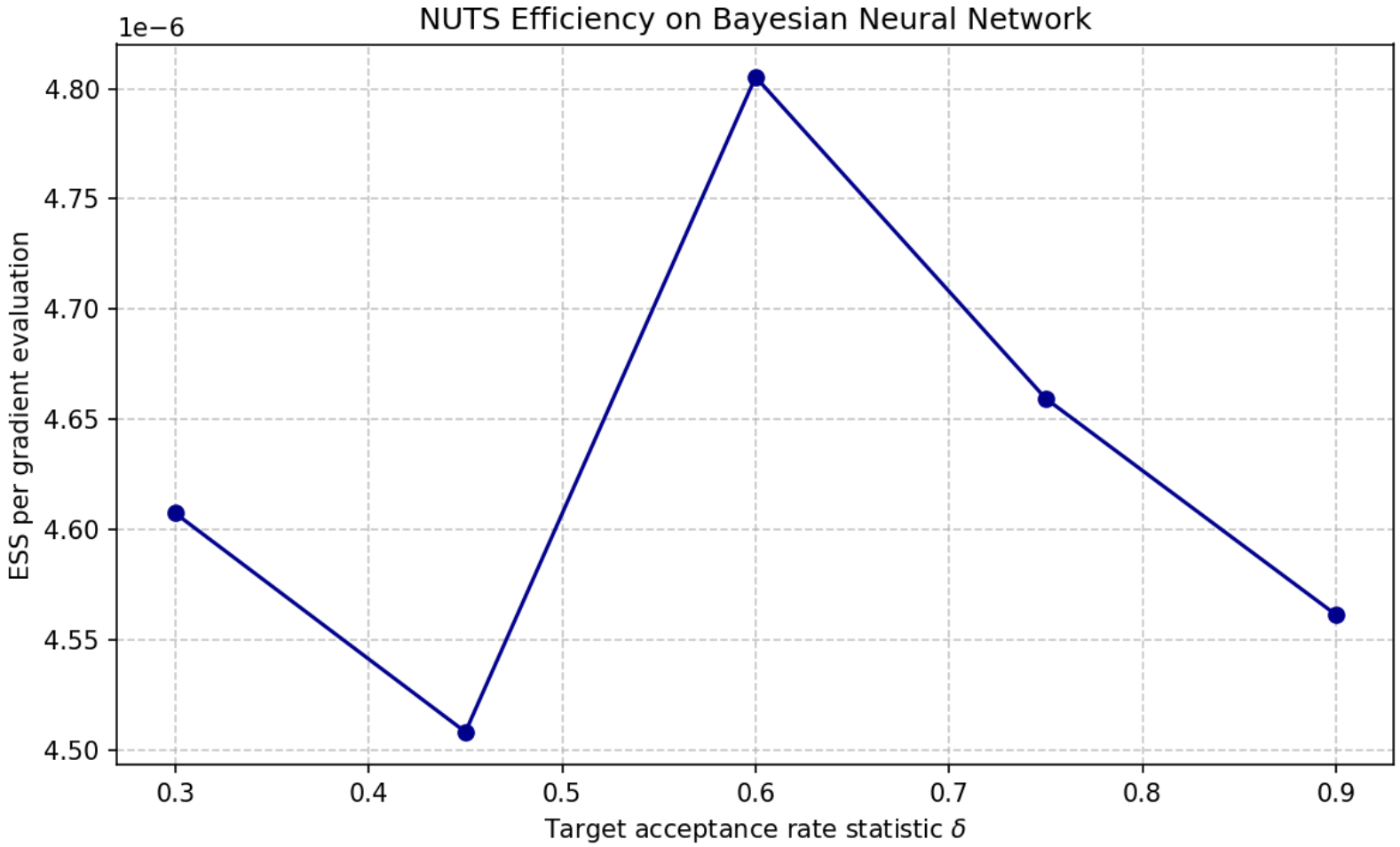

Figure: ESS/gradient comparison on the BNN. NUTS achieves higher effective samples per gradient evaluation than HMC across all tested configurations.

Stress Test 2: GP Classification (150-D)

Setup:

- GP classification with RBF kernel ($\ell = 1$, $\sigma_f^2 = 1$)
- 150 points from the California Housing dataset
- Fully dense 150×150 covariance matrix K
- Gradient $\nabla \log p(f) = -K^{-1} f + y \odot \sigma(-y \odot f)$ requires a dense solve at every leapfrog step

Configuration: $M = 500$, $M_{\text{adapt}} = 250$

NUTS achieves 2.5×10^{-3} ESS/grad at $\delta = 0.7$, requiring fewer gradient evaluations than HMC for equivalent effective samples.

GPC: Kernel Matrix and Posterior Correlations

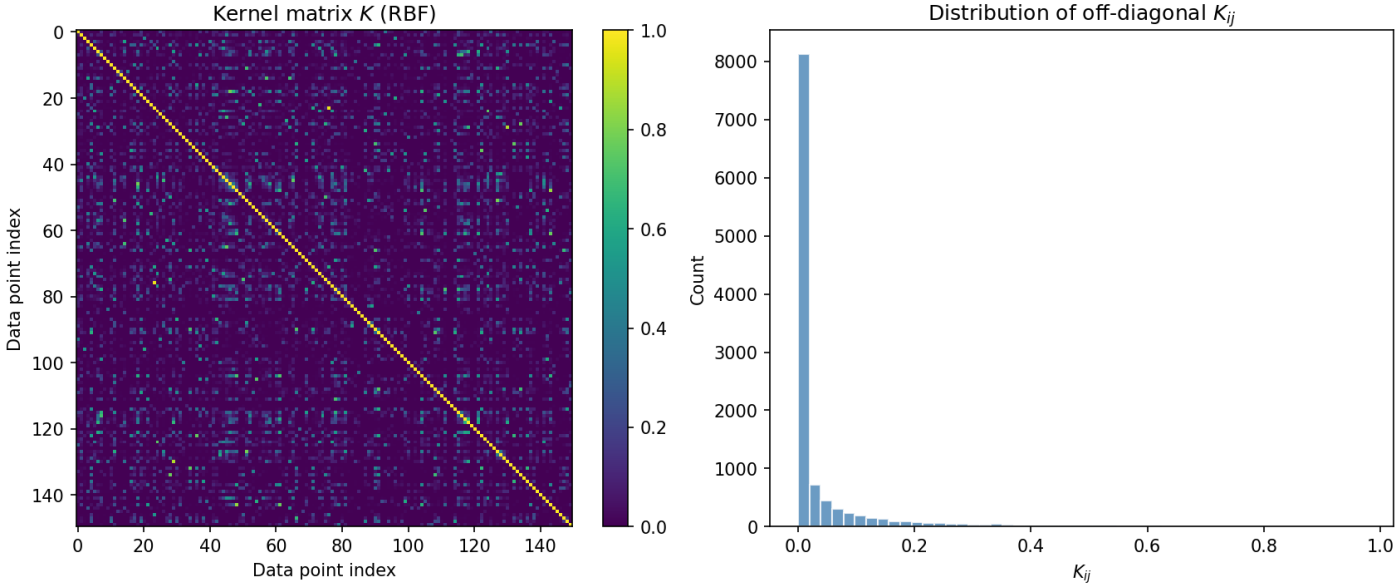

Figure: RBF kernel matrix K exhibiting dense correlation structure.

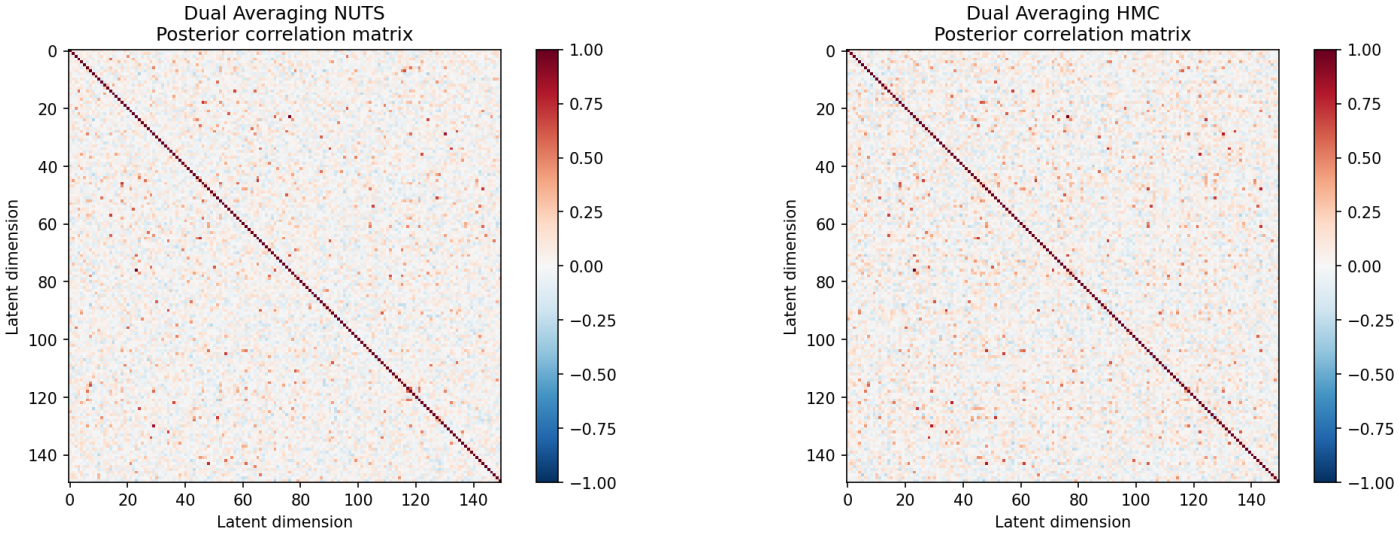

Figure: Posterior correlations among the latent function values f.

GPC: Trajectory Lengths and Efficiency

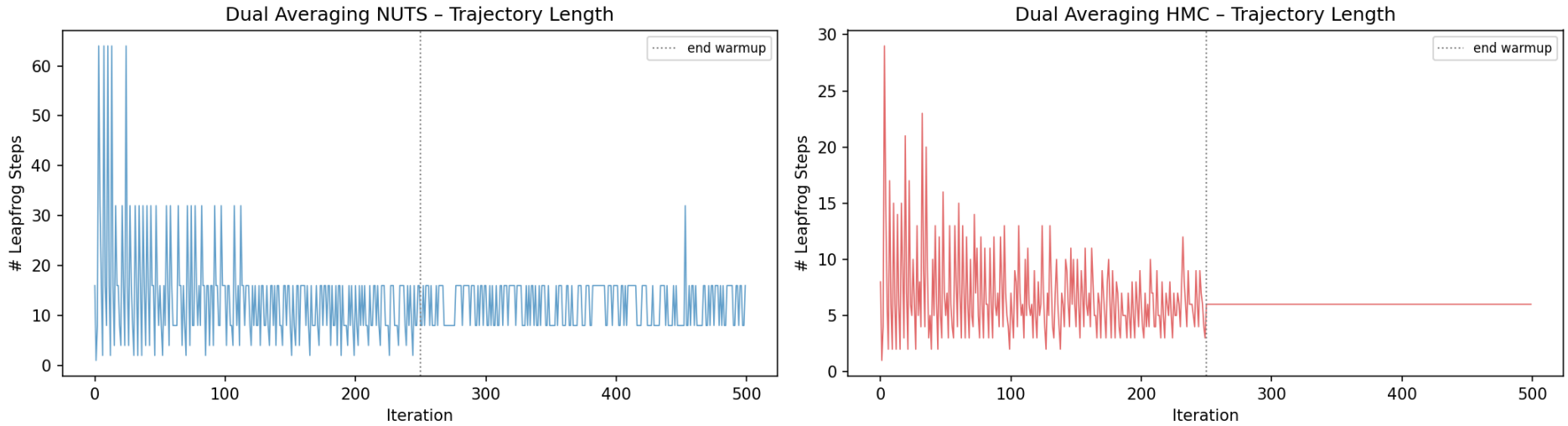

Figure: NUTS trajectory lengths on the GP classification posterior. The dense correlation structure demands longer trajectories to traverse the strongly coupled latent space.

GPC: Efficiency Comparison

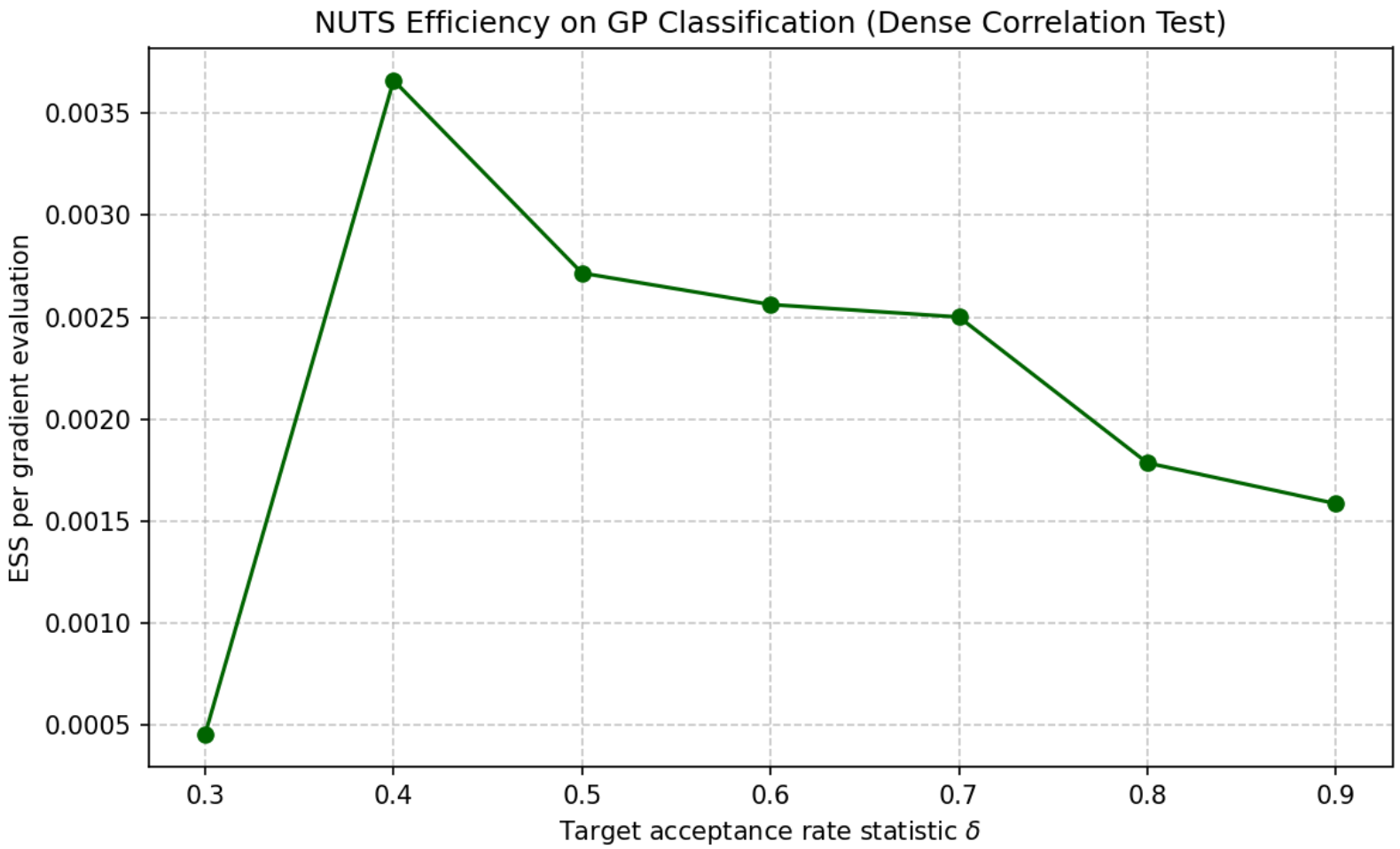

Figure: ESS/gradient on GP classification. NUTS requires fewer gradient evaluations than HMC for equivalent effective samples under dense correlations.

Stress Test 3: Neal's Funnel (10-D)

Model:

$$v \sim \mathcal{N}(0, 9), \qquad x_i \mid v \sim \mathcal{N}(0, e^v), \quad i = 1, \ldots, 9$$

Challenge:

- As $v \to -\infty$: conditional variance $e^v \to 0$ creates a microscopic neck demanding tiny steps
- Wide top ($v > 0$) requires large steps
- HMC's fixed step size cannot handle both regimes simultaneously

This is a canonical pathological target [Nea03] that exposes the limitations of fixed-trajectory samplers.

Funnel: Scatter Plots

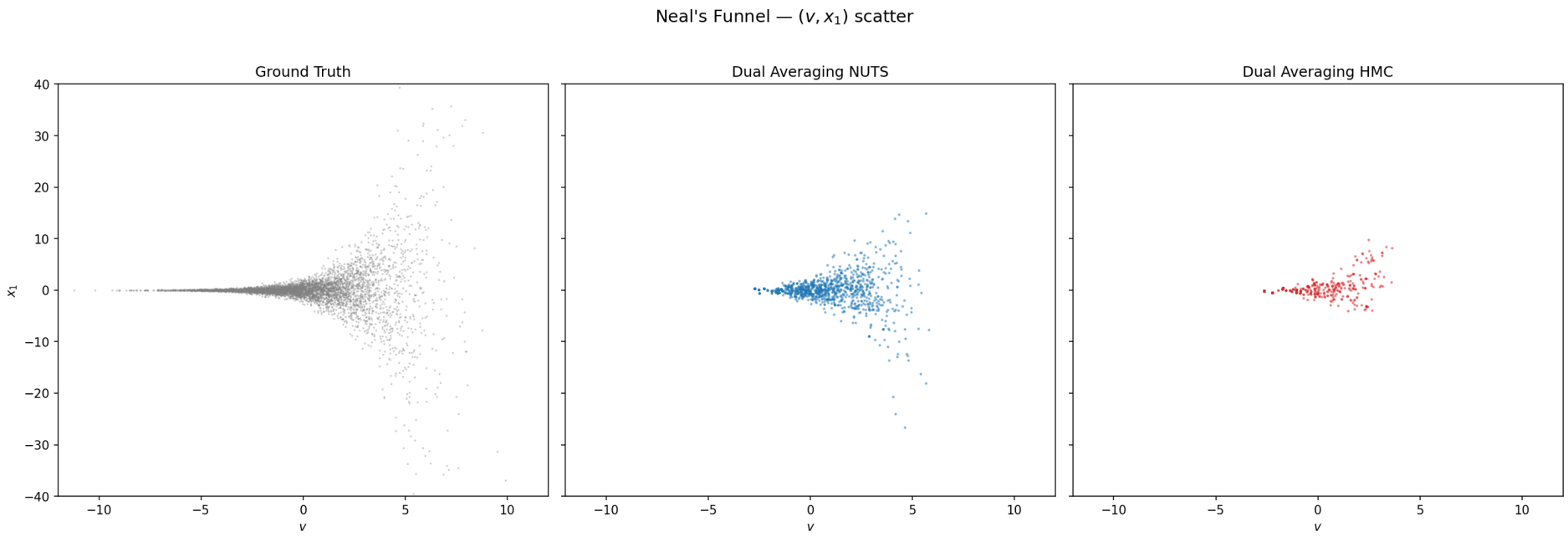

Figure: Neal's Funnel (v, x_1) scatter. **Left:** ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to enter the narrow neck ($v < 0$), producing biased samples.

Funnel: Trace Plots

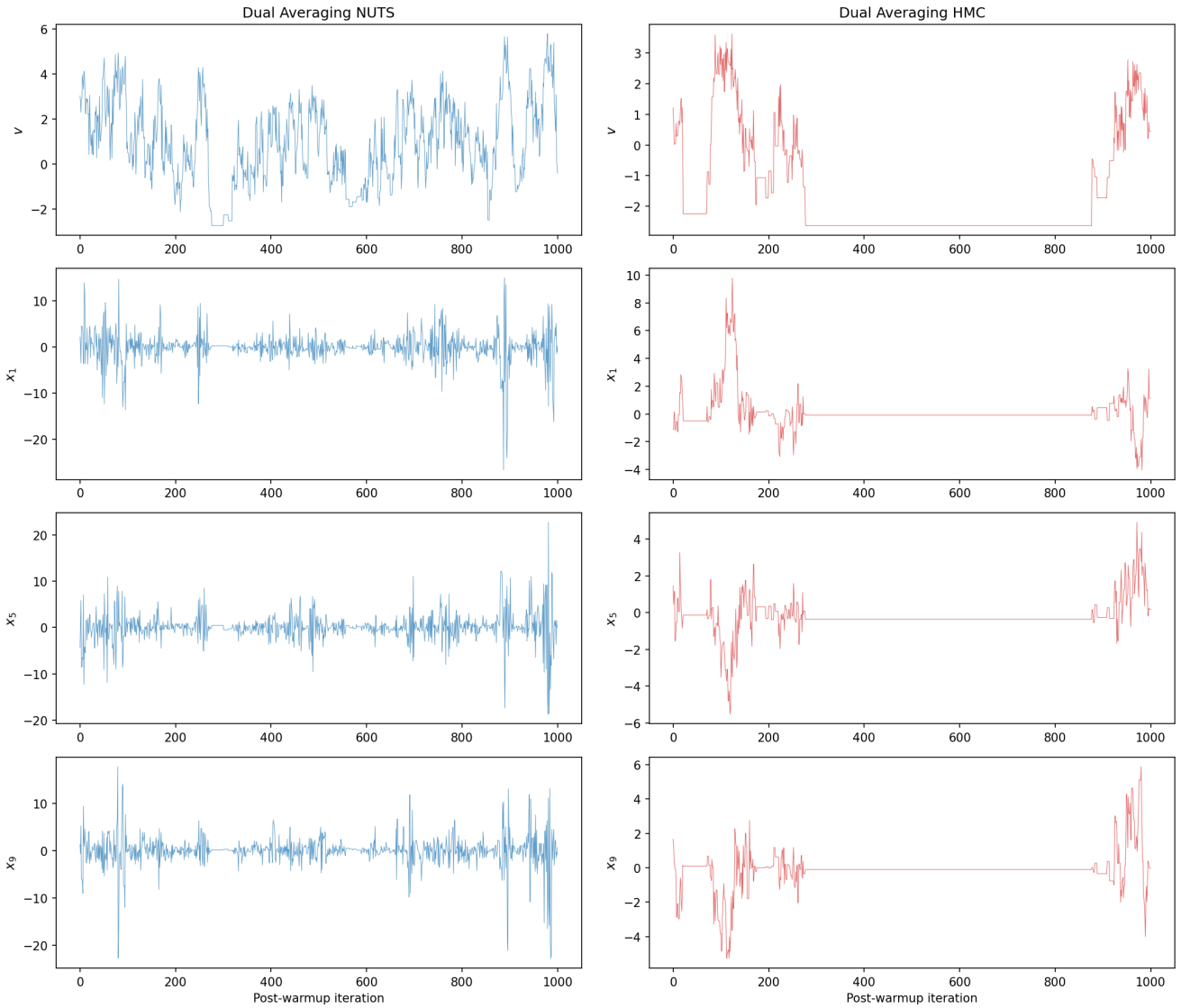

Figure: Trace plots on Neal's Funnel. NUTS transitions freely across the full v range with good mixing. HMC gets trapped with flat segments indicating rejected proposals.

Funnel: Marginal Distribution of v

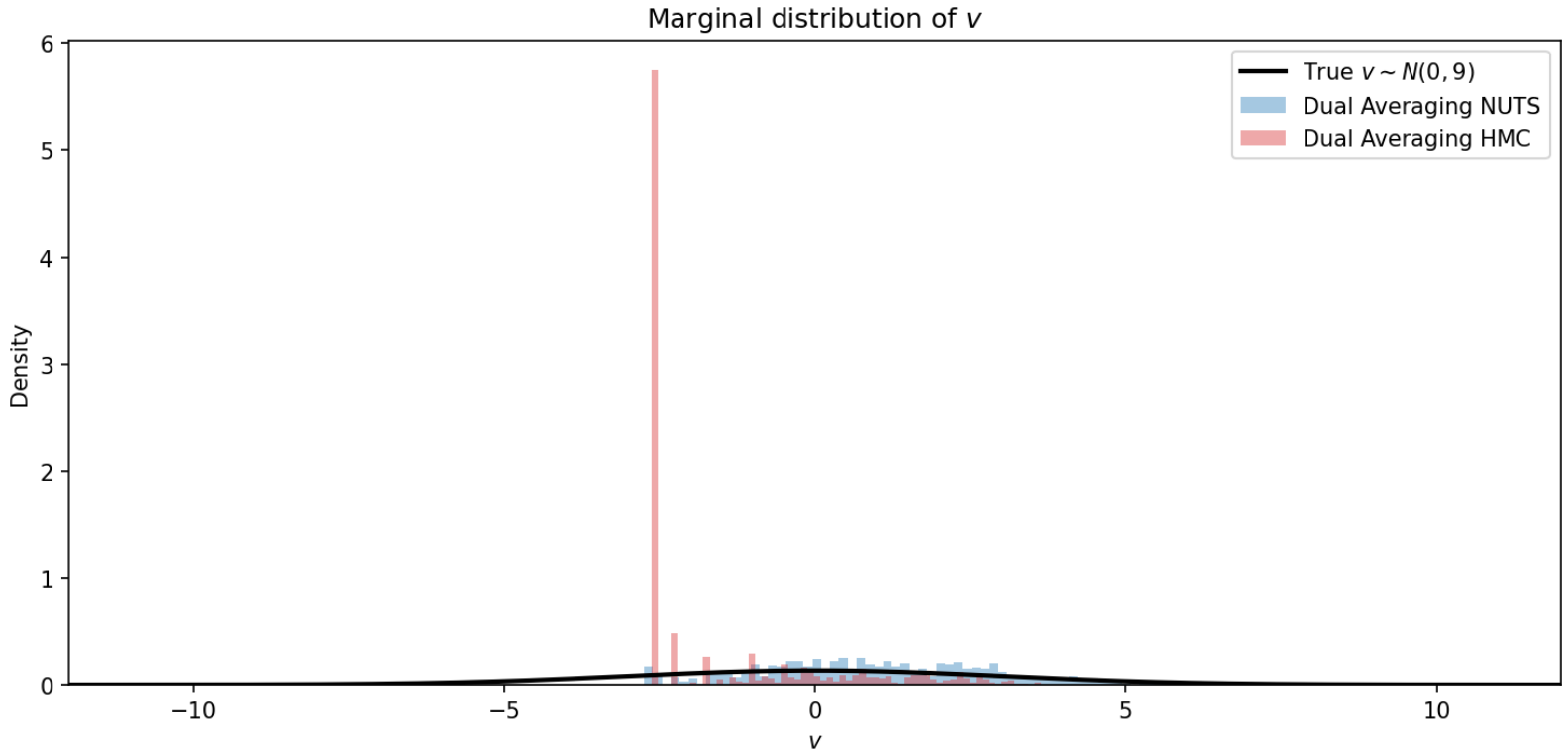

Figure: Marginal distribution of v. NUTS recovers $\mathrm{Var}(v) \approx 9$ (the true value), while HMC severely underestimates it, confined to $v \in [0, 4]$.

Funnel: Efficiency Comparison

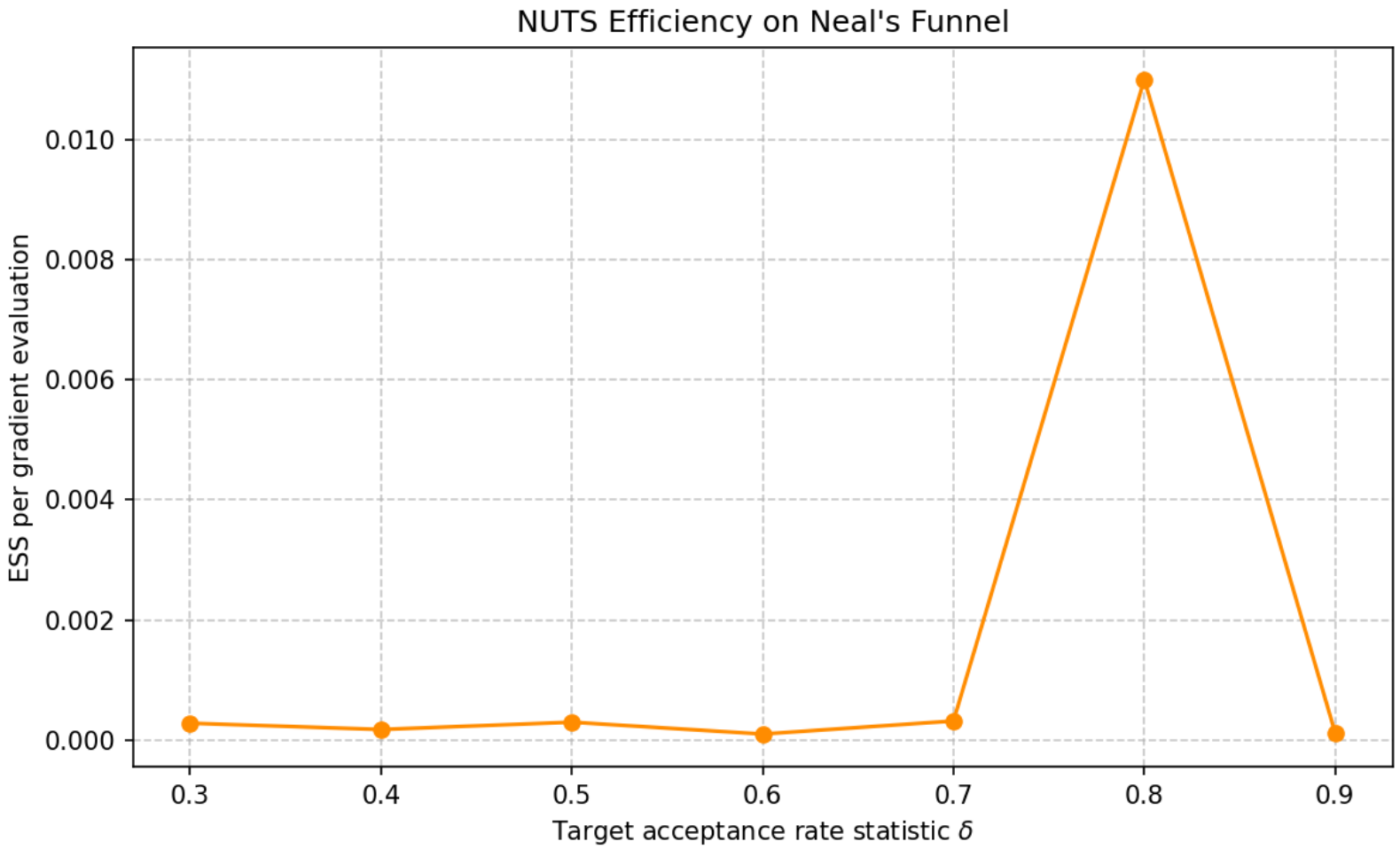

Figure: ESS/gradient on Neal's Funnel. The gains are most dramatic when the posterior geometry varies spatially.

Summary

Strengths of the article:

- Resolves a central practical limitation of HMC through a conceptually simple yet theoretically rigorous construction
- No-U-Turn criterion and recursive doubling preserve detailed balance
- Dual averaging for step-size adaptation makes NUTS essentially tuning-free

Limitations:

- Dual-averaging hyperparameters justified empirically, not theoretically
- Performance guarantees are mainly asymptotic
- Original experiments cover a limited range of posterior geometries

Key Takeaways

Our implementation validates the original results: NUTS matches or exceeds the best-tuned HMC on all four benchmarks while eliminating trajectory-length tuning.

Our stress tests demonstrate that adaptive trajectory length is critical on posteriors with complex geometries:

- **BNN:** NUTS navigates 14,400 equivalent modes; HMC stagnates
- **GPC:** NUTS handles dense 150×150 correlations efficiently
- **Neal's Funnel:** NUTS covers the full funnel; HMC fails catastrophically

NUTS is confirmed as a suitable **general-purpose inference engine** for modern Bayesian models.

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[Nea+11] Radford M Neal et al. "MCMC using Hamiltonian dynamics". In: *Handbook of markov chain monte carlo* 2.11 (2011), p. 2.

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Appendix: BNN Adaptation Diagnostics

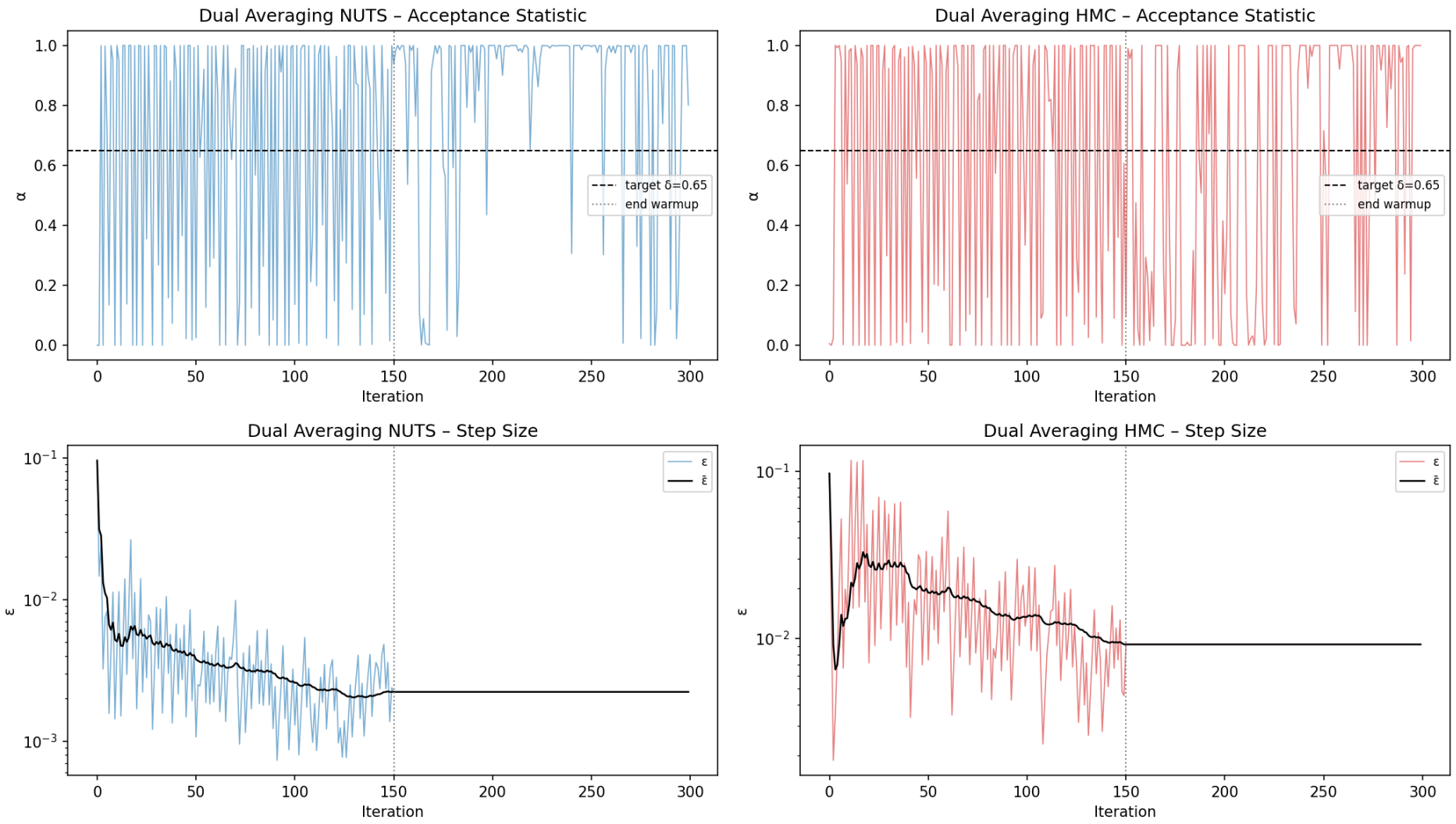

Figure: Dual-averaging convergence on the BNN posterior. The step size $\bar{\varepsilon}$ stabilises within the warm-up phase.

Appendix: GPC Trace Plots

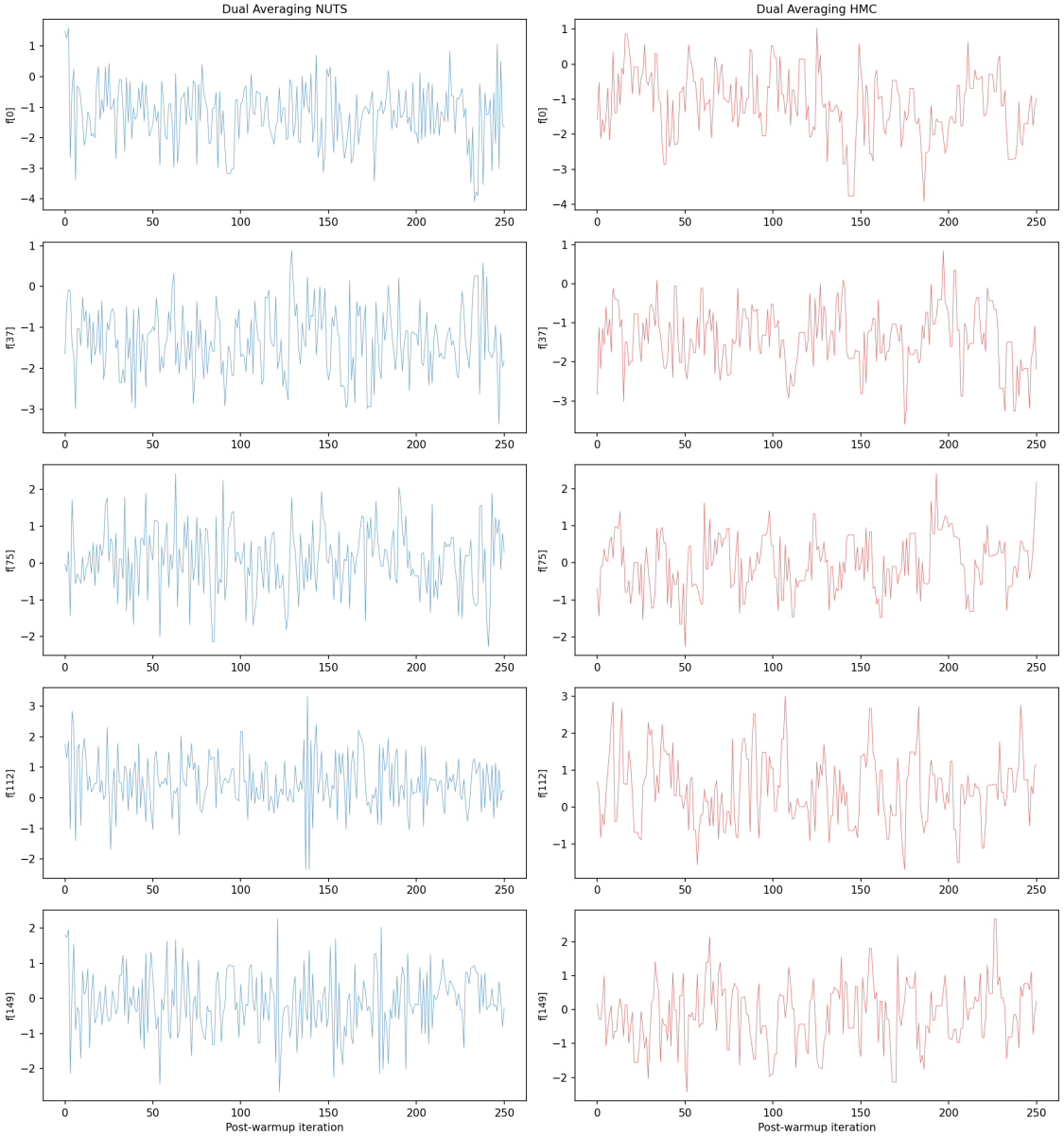

Figure: Trace plots for selected latent function values on the GP classification posterior.

Appendix: Funnel Trajectory Lengths

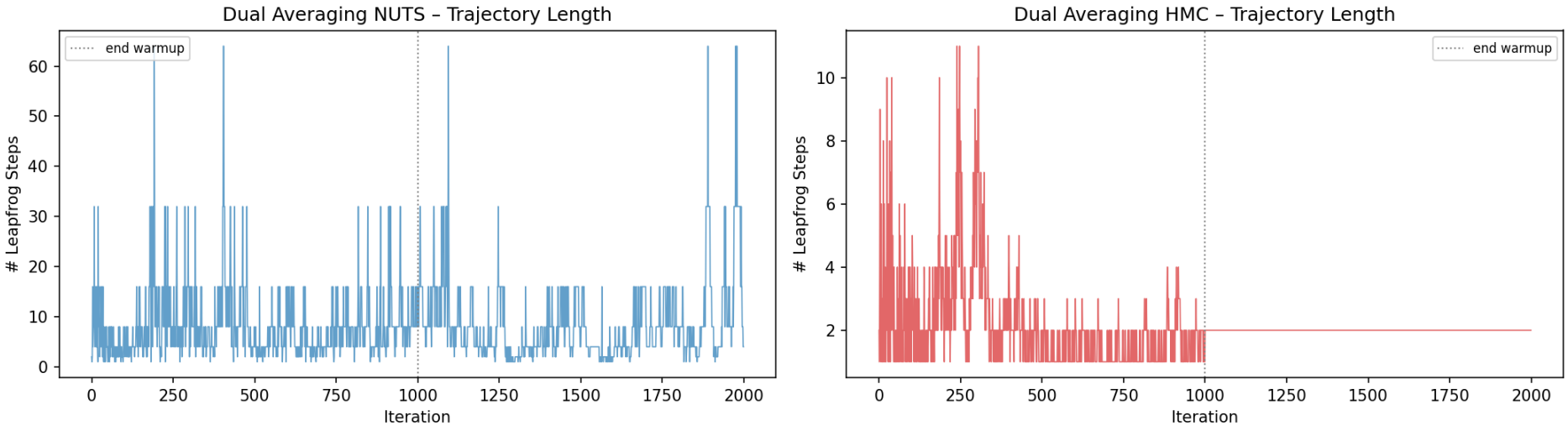

Figure: NUTS trajectory lengths on Neal's Funnel. The adaptive mechanism adjusts trajectory length to the local geometry.